

# SHTUKAS

## 1. SHTUKAS

First, I want to motivate the concept of a shtuka from a number theory perspective, following Gaitsgory's account [1]. In order to get a some sort of map from the automorphic side to the spectral side, the way of doing this over  $\mathbf{Q}$  is to look at schemes over number fields which have both an action of  $G(\mathbf{A}_f)$  and an action of  $G_K$  for a number field  $K$ . More specifically, we need some specific control over these actions, which comes in the form of relating the action of the Hecke algebra  $\mathbf{T}$  to the Galois action of  $\text{Frob}_p \in G_K$ .

This can also be done in the function field setting, where the resulting objects are a great deal simpler. We have a correspondence

$$\begin{array}{ccc} & \text{Hecke}_{X^I} & \\ \swarrow & & \searrow \\ \text{Bun}_G & & \text{Bun}_G \end{array}$$

which gives an action of Hecke operators on  $G$ -bundles. In particular, Hecke operators act by modifications of  $G$ -bundles.

Since the Hecke stack has a map  $\pi : \text{Hecke}_{X^I} \rightarrow X^I$ , one idea might be to use  $\pi_!$  in some way to produce a sheaf on  $X^I$  with an action of the Hecke algebra. The naive expectation might be that we can get a Galois action via  $\pi_1^{\text{et}}(X)^I$ , but actually this fails quite badly: there is a difference between  $\pi_1^{\text{et}}(X^I)$  and  $\pi_1^{\text{et}}(X)^I$ , for example take  $X = \mathbf{A}_{\mathbb{F}_p}^1$ . What is needed to fix this issue is the idea of a partial Frobenius: if we add compatibility with partial Frobenii on  $X^I$ , Drinfeld's lemma gives us an actual Galois action. There is not too much hope in this situation because the action of Frobenius on a  $G$ -bundle and modifications of  $G$ -bundles have nothing to do with each other, so we shouldn't expect any such compatibility.

We define a stack  $\text{Sht}_{G,I}$  as a fiber product

$$\begin{array}{ccc} \text{Sht}_I & \longrightarrow & \text{Hecke}_{X^I} \\ \downarrow & & \downarrow \bar{h}, \bar{h} \\ \text{Bun}_G & \xrightarrow{\text{Frob}, \text{Id}} & \text{Bun}_G \times \text{Bun}_G \end{array}$$

so that we've forced modifications of  $G$ -bundles and Frobenius to have something to do with each other. This comes equipped with a map  $\pi' : \text{Sht}_I \rightarrow X^I$  via the Hecke stack. As a basic example, we know that

$$\text{Funct}_c(\text{Bun}_G(\mathbb{F}_q)) = \text{Autom}$$

which clearly has a Hecke action. This is coming from  $\text{Sht}_\emptyset$ : we have a cartesian square

$$\begin{array}{ccc}
\mathrm{Sht}_\emptyset = \mathrm{Bun}_G(\mathbf{F}_q) & \longrightarrow & \mathrm{Bun}_G \\
\downarrow & & \downarrow \\
\mathrm{Bun}_G & \xrightarrow{\mathrm{Frob}, \mathrm{Id}} & \mathrm{Bun}_G \times \mathrm{Bun}_G.
\end{array}$$

Taking  $\pi_!^* \overline{\mathbf{Q}}_\ell$ , we get  $\mathrm{Autom}$ .

In general, we use the sheaves  $\mathcal{S}_V$  on the Hecke stack coming from classical geometric Satake. Namely, for such a  $V \in \mathrm{Rep}(\check{G})^{\otimes 1}$  we obtain a sheaf  $\mathcal{S}_V \in \mathrm{Shv}(\mathrm{Hecke}_{X^1})$ . This induces a sheaf  $\mathcal{S}'_V$  on  $\mathrm{Sht}_1$ , and we then define

$$\mathrm{Sht}_{1,V} := \pi_1^* \mathcal{S}'_V \in \mathrm{Shv}(X^1).$$

It is now known by Cong Xue's smoothness theorem that these lie in  $\mathbf{QLisse}(X^1)$  [2]. Furthermore, they now come equipped with equivariance with partial Frobenii on  $X^1$  so that we actually get a Galois action.

The sheaves  $\mathrm{Sht}_{1,V}$  can be assembled into a sheaf  $\mathrm{DrinfSht}$  in  $\mathbf{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}})$ . There is an alternative construction  $\mathrm{Drinf}$  of this sheaf as a categorical trace, which we'll talk about.

These are really the main player in everything done in global function field Langlands. By adding level structures, these are essential in the proof for  $\mathrm{GL}_n$ . Lafforgue utilizes these, again with level structure, to construct *excursion operators* [3]. These correspond to generators of an algebra  $\mathcal{B}$  related to  $\mathcal{E}xc = \Gamma(\mathrm{LocSys}^{\mathrm{arithm}}, \mathcal{O})$  (the discrete, underived version of this). This algebra acts on automorphic forms via the excursion operators, and then automorphic forms get decomposed into  $\mathcal{B}$ -eigenspaces. By construction, the characters of this algebra correspond to  $L$ -parameters, and compatibility of excursion and Hecke operators gives the desired compatibility of the decomposition

$$C_c^{\mathrm{cusp}}(\mathrm{Bun}_{G,N}(\mathbf{F}_q)/\Xi, \overline{\mathbf{Q}}_\ell) = \bigoplus_{\sigma: \mathrm{Gal}(\overline{F}/F) \rightarrow \check{G}(\overline{\mathbf{Q}}_\ell)} \mathfrak{h}_\sigma$$

with Satake parameters. The sheaf  $\mathrm{DrinfSht} \simeq \mathrm{Drinf}$  captures an unramified version of this via the action of  $\mathcal{E}xc$  acting on its global sections, which are going to be  $\mathrm{Autom}$ .

## 2. CONSTRUCTION OF DRINF SHT

Our first task is to assemble these elements of  $\mathbf{QLisse}(X^1)$  into a single object  $\mathrm{DrinfSht} \in \mathbf{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}})$ .

**LEMMA 2.1.** The category  $\mathbf{QCoh}(\mathrm{LocSys}^{\mathrm{restr}})$  is *canonically* self-dual, induced by the pairing

$$\mathcal{F}_1, \mathcal{F}_2 \mapsto \Gamma_!(\mathrm{LocSys}^{\mathrm{restr}}, \mathcal{F}_1 \otimes \mathcal{F}_2).$$

The same holds for  $\mathrm{LocSys}^{\mathrm{arithm}}$ , but with  $\Gamma$ .

*Proof.* This holds using  $\Gamma$  for any quasicompact stack which is locally almost of finite type. When we are in a situation where things are not quasicompact, we make a replacement  $\Gamma_!$  to still get a duality.  $\square$

Using the canonical self-duality, we will be able to construct objects from collections of functors resembling shtukas. Some definitions are in order first.

Let  $I \in \mathbf{FinSet}$ , and  $V \in \mathrm{Rep}(\check{G})^{\otimes I}$ . We will produce an object

$$\mathcal{E}_V^I \in \mathbf{QCoh}(\mathrm{LocSys}^{\mathrm{restr}}) \otimes \mathbf{QLisse}(X)^{\otimes I}.$$

Let us explain how this is produced in general. If  $\mathcal{H}$  is a dualizable gentle Tannakian category, we look at the map

$$\mathrm{Rep}(\check{G})^{\otimes I} \rightarrow \mathrm{QCoh}(\mathrm{Maps}(\mathrm{Rep}(\check{G}), \mathcal{H}))^{\otimes I} \otimes \mathcal{H}^{\otimes I}$$

and applying the tensor product functor we land in  $\mathrm{QCoh}(\mathrm{Maps}(\mathrm{Rep}(\check{G}), \mathcal{H})) \otimes \mathcal{H}^{\otimes I}$ .

We obtain the first functor by passing to the limit on the functors

$$\mathrm{Rep}(\check{G}) \rightarrow \mathrm{QCoh}(S) \otimes \mathcal{H}$$

for  $S \in \mathrm{Sch}_{/\mathrm{Maps}(\mathrm{Rep}(\check{G}), \mathcal{H})}^{\mathrm{aff}}$ , which will be  $\mathrm{QCoh}(\mathrm{Maps}(\mathrm{Rep}(\check{G}), \mathcal{H})) \otimes \mathcal{H}$  by virtue of  $\mathcal{H}$  being dualizable.

Now apply this formalism for  $\mathrm{QLisse}$ , and we have explained how to produce  $\mathcal{E}^I$ . As constructed, this is a functor and not an object;  $\mathcal{E}_V^I$  is the value on  $V$ .

**THEOREM 2.2.** The functor

$$\mathrm{coLoc} : \mathrm{QCoh}(\mathrm{LocSys}^{\mathrm{restr}}) \rightarrow \mathrm{Maps}(\mathrm{Rep}(\check{G})^{\otimes \mathrm{FinSet}}, \mathrm{QLisse}(X)^{\otimes \mathrm{FinSet}})$$

is an equivalence.

This sends  $\mathcal{F}$  to the functors

$$V \mapsto (\Gamma_!(\mathrm{LocSys}^{\mathrm{restr}}, -) \otimes \mathrm{Id})(\mathcal{E}_V^I \otimes \mathcal{F})$$

from  $\mathrm{Rep}(\check{G})^{\otimes I} \rightarrow \mathrm{QLisse}(X)^{\otimes I}$ .

In particular, we can take the collection of such functors given by cohomology of shtukas. This produces the desired object. There is some sleight of hand here: the actual objects we got lived in  $\mathrm{QLisse}(X^I)$ , not  $\mathrm{QLisse}(X)^{\otimes I}$ . Fortunately for us, the Künneth map  $\mathrm{QLisse}(X)^{\otimes I} \rightarrow \mathrm{QLisse}(X^I)$  is an equivalence.

We are not yet done: we have a quasicohherent sheaf on the wrong stack. We need to figure out how to descend these.

**LEMMA 2.3.** The datum of exhibiting  $\mathcal{F}$  as  $i_* \mathcal{F}^{\mathrm{arithm}}$  for some  $\mathcal{F}^{\mathrm{arithm}} \in \mathrm{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}})$ , where  $i : \mathrm{LocSys}^{\mathrm{arithm}} \rightarrow \mathrm{LocSys}^{\mathrm{restr}}$  is the natural map, is equivalent to a compatible structure of partial Frobenii on the functors

$$V \mapsto (\Gamma_!(\mathrm{LocSys}^{\mathrm{restr}}, -) \otimes \mathrm{Id})(\mathcal{E}_V^I \otimes \mathcal{F})$$

from  $\mathrm{Rep}(\check{G})^{\otimes I} \rightarrow \mathrm{QLisse}(X)^{\otimes I}$ .

By construction, we took these to arise from  $\mathrm{Sht}_{I,V}$ . These are equipped with partial Frobenii (as we know from number theory), so we actually get a sheaf  $\mathrm{DrinfSht}$  on  $\mathrm{LocSys}^{\mathrm{arithm}}$ .

This sheaf acts as a sort of universal shtuka, as it assembles all shtukas into a single object and allows us to recover them through the tautological objects  $\mathcal{E}_V^I$ . Moreover, note that if I put  $I = \emptyset$  and  $V$  as the trivial representation, I get  $\mathrm{Funct}_c(\mathrm{Bun}_G(\mathbf{F}_q)) = \mathrm{Autom}$ . In particular, global sections of this sheaf give us unramified automorphic forms.

## 3. TRACE AND DRINF

Let us now define another sheaf, called **Drinf**. We will define objects

$$\widetilde{\text{Sht}}_{I,V} \in \text{QLisse}(X^I)$$

by considering the Hecke functors

$$H(V, -) : \text{Shv}_{\text{Nilp}}(\text{Bun}_G) \rightarrow \text{Shv}_{\text{Nilp}}(\text{Bun}_G) \otimes \text{QLisse}(X^I).$$

Now precomposing with  $\text{Frob}_*$ , we obtain a functor

$$F : \mathbf{C} \rightarrow \mathbf{C} \otimes \mathbf{D}$$

where  $\mathbf{C} = \text{Shv}_{\text{Nilp}} \text{Bun}_G$  and  $\mathbf{D} = \text{QLisse}(X^I)$ . As  $\text{Shv}_{\text{Nilp}}(\text{Bun}_G)$  is dualizable, we obtain a categorical trace of this functor  $F$  as

$$\text{Vect}_{\overline{\mathbb{Q}}_\ell} \rightarrow \mathbf{C}^\vee \otimes \mathbf{C} \rightarrow \mathbf{C}^\vee \otimes \mathbf{C} \otimes \mathbf{D} \rightarrow \mathbf{D}.$$

This construction results in  $\widetilde{\text{Sht}}_{I,V} \in \text{QLisse}(X^I)$ .

This produces a sheaf **Drinf** on  $\text{LocSys}^{\text{restr}}$ . However, if we want to see it actually descends to  $\text{LocSys}^{\text{arithm}}$ , it is better construct it directly as an enhanced trace.

We define

$$\text{Drinf} := \text{Tr}_{\text{Frob}^*, \text{QCoh}(\text{LocSys}^{\text{restr}})}^{\text{enh}}(\text{Frob}_*, \text{Shv}_{\text{Nilp}}(\text{Bun}_G)).$$

This is a 2-categorical construction called enhanced trace, where we are regarding  $\text{Shv}_{\text{Nilp}}(\text{Bun}_G)$  as a module category over  $\text{QCoh}(\text{LocSys}^{\text{restr}})$ . First, to the pair  $(\text{Frob}^*, \text{QCoh}(\text{LocSys}^{\text{restr}}))$  we associate the trace

$$\text{Tr}(\text{Frob}^*, \text{QCoh}(\text{LocSys}^{\text{restr}})) \simeq \text{QCoh}(\text{LocSys}^{\text{arithm}}).$$

Let me explain what this means. In general, given a symmetric monoidal category  $\mathbf{A}$  with endofunctor  $F_A$ , define

$$\text{Tr}(F_A, \mathbf{A}) := F_A \otimes_{\mathbf{A} \otimes \mathbf{A}^{\text{op}}} \mathbf{A}$$

where we view the endofunctor  $F_A$  as an  $\mathbf{A} \otimes \mathbf{A}^{\text{op}}$ -bimodule category (precisely,  $\mathbf{A}$  with an action on one side twisted by  $F_A$ ). For the previous pair, the result of this construction is  $\text{QCoh}(\text{LocSys}^{\text{arithm}})$ .

Now to the pair  $(\text{Frob}_*, \text{Shv}_{\text{Nilp}}(\text{Bun}_G))$ , using that  $\text{Shv}_{\text{Nilp}}(\text{Bun}_G)$  is a module category over  $\text{QCoh}(\text{LocSys}^{\text{restr}})$ , we can associate a class in this Hochschild homology which is going to be **Drinf**.

For the construction of this class, let us move to the general setting where we have pairs  $(\mathbf{A}, F_A)$  and  $(\mathbf{M}, F_M)$  where  $\mathbf{M}$  is a dualizable module category over a rigid symmetric monoidal category  $\mathbf{A}$  and the second part of the pairs is an endofunctor (which is symmetric monoidal in the case of  $\mathbf{A}$ ). We ask for a compatibility datum, namely that the diagram

$$\begin{array}{ccc} \mathbf{A} \otimes \mathbf{M} & \longrightarrow & \mathbf{M} \\ \downarrow F_A \otimes F_M & & \downarrow F_M \\ \mathbf{A} \otimes \mathbf{M} & \longrightarrow & \mathbf{M} \end{array}$$

commutes. Viewing a module category as a 1-morphism

$$T : \text{DGCat} \rightarrow \mathbf{A} - \text{Mod}$$

this data is a natural transformation

$$\alpha : T \circ \text{id} \rightarrow F_A \circ T.$$

In such a situation, categorical constructions provide us with a morphism

$$\mathrm{Tr}(\mathrm{id}, \mathbf{DGCat}) \rightarrow \mathrm{Tr}(F_A, \mathbf{A} - \mathbf{Mod}).$$

That is, a morphism from  $\mathbf{Vect} \rightarrow \mathrm{Tr}(F_A, \mathbf{A} - \mathbf{Mod})$ ; we get an object.

With our given pairs, this is the class in  $\mathrm{Tr}(\mathrm{Frob}^*, \mathbf{QCoh}(\mathrm{LocSys}^{\mathrm{restr}})) \simeq \mathbf{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}})$  we want to associate.

One can prove that we recover  $\widetilde{\mathrm{Sht}}_{I,V}$  in the following way.

**THEOREM 3.1 (Gaitsgory, [1]).** The sheaf  $\mathbf{Drinf} \in \mathbf{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}})$  has the property

$$(\Gamma(\mathrm{LocSys}_{\check{G}}^{\mathrm{arithm}}, -) \otimes \mathrm{Id})(\mathbf{Drinf} \otimes \mathcal{E}_V^I) \simeq \widetilde{\mathrm{Sht}}_{I,V}$$

Here, by abuse of notation we write  $\mathcal{E}_V^I$  for the restriction to  $\mathrm{LocSys}^{\mathrm{arithm}}$ . This result shows that these sheaves in  $\mathbf{QLisse}(X^I)$  are equipped with partial Frobenii, which means that if we had done the original construction we would be able to descend it to  $\mathrm{LocSys}^{\mathrm{arithm}}$ . Note here that we also use  $\Gamma$ , as here the stack ends up being quasicompact (although this is nontrivial to show).

#### 4. A (FIXABLE) FAKE PROOF

The trace-conjecture statement used in Gaitsgory's account is:

**CONJECTURE 4.1 (Trace conjecture, [1]).** There is a canonical isomorphism

$$\mathbf{Drinf} \simeq \mathbf{DrinfSht}$$

in  $\mathbf{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}})$ .

**REMARK 4.2.** Once we have this isomorphism, using the tautological objects  $\mathcal{E}_V^I$  on both must produce the same results. We know that  $\mathbf{Drinf}$  gives  $\mathrm{Sht}_{I,V}$ , and for  $\mathbf{DrinfSht}$  this gives  $\mathrm{Sht}_{I,V}$ . We know we get  $\mathrm{Sht}_{I,V}$  on  $\mathrm{LocSys}^{\mathrm{restr}}$ , and the partial Frobenii descend it to  $\mathrm{LocSys}^{\mathrm{arithm}}$  and  $\mathcal{E}_V^{I,\mathrm{arithm}}$  extracts the same element of  $\mathbf{QLisse}(X^I)$ .

The canonical identification of these gives the trace conjecture.

This is a nontrivial result, but I'd like to explain why you should believe these quasicoherent sheaves on  $\mathrm{LocSys}^{\mathrm{arithm}}$  are isomorphic following Gaitsgory's argument.

This argument crucially assumes we live in a simpler world that allows

$$\mathbf{Shv}(\mathrm{Bun}_G \times \mathrm{Bun}_G) \simeq \mathbf{Shv}(\mathrm{Bun}_G) \otimes \mathbf{Shv}(\mathrm{Bun}_G).$$

This is not actually true, and one needs to use  $\mathbf{Shv}_{\mathrm{Nilp}}$  instead to make this work. The correct argument also uses a self-duality of  $\mathbf{Shv}_{\mathrm{Nilp}}(\mathrm{Bun}_G)$  instead of  $\mathbf{Shv}(\mathrm{Bun}_G)$ , which is also more difficult. Both of these results need Beilinson's spectral projector to be done correctly.

We will pretend the following are true:

- $\mathbf{Shv}(\mathrm{Bun}_G \times \mathrm{Bun}_G) \simeq \mathbf{Shv}(\mathrm{Bun}_G) \otimes \mathbf{Shv}(\mathrm{Bun}_G)$ .

- $\text{LocSys}^{\text{restr}}$  is a quasicompact algebraic stack (I think we don't actually need this; it's just to avoid  $\Gamma_!$  for the self-duality). In Gaitsgory's argument he actually pretends that all étale local systems assemble into an algebraic stack, but we don't need to go this far in our assumptions.
- Redefine  $\text{Drinf} := \text{Tr}_{\text{QCoh}(\text{LocSys}^{\text{restr}})}^{\text{enh}}(\text{Frob}^!, \text{Shv}(\text{Bun}_G))$ , so we can still get compactly supported functions on  $\text{Bun}_G(\mathbf{F}_q)$  on global sections. In particular, we want to get

$$\text{Tr}(\text{Frob}^!, \text{Shv}(\text{Bun}_G)) = \text{Funct}_c(\text{Bun}_G(\mathbf{F}_q))$$

for the usual trace.

The argument I'll explain follows essentially the same overall structure as the correct one, just with less technical details as we are allowed to make these assumptions.

The basic idea is that both  $\text{Drinf}$  and  $\text{DrinfSht}$  satisfy

$$\Gamma(\text{LocSys}^{\text{arithm}}, \mathcal{E}_{V,x} \otimes \mathcal{F}) \simeq H^\bullet(\text{Bun}_G \times_{\text{Frob}_*, \text{Bun}_G \times \text{Bun}_G} \text{Hecke}_x, \mathcal{S}'_V).$$

For  $\text{DrinfSht}$ , this is by definition in our setting.

Here,  $\mathcal{E}_{V,x}$  is originally produced on  $\text{LocSys}^{\text{restr}}$  by using the inclusion of a rational point  $x \rightarrow X$ , and the representation  $V$  of  $\check{G}$  gives us an element in  $\text{QCoh}$  for  $\check{G}$ -local systems on  $x$ . We then restrict to get the desired sheaf. The sheaf  $\mathcal{S}'_V$  is given by geometric Satake, and originally lives on the Hecke stack at  $x$  before we pull it back to the fiber product.

This property uniquely characterizes a sheaf in  $\text{QCoh}(\text{LocSys}^{\text{restr}})$ . Indeed, given any sheaf the functor  $\Gamma(\text{LocSys}^{\text{arithm}}, - \otimes \mathcal{F})$  gives a functor to  $\mathbf{Vect}$ . Then canonical self-duality gives us a corresponding object, which is then  $\mathcal{F}$ . Since  $\mathcal{E}_{V,x}$  generate (by this I mean everything can be written as colimits involving them, which suffices if we want a cocontinuous functor to use duality on), this describes the entire functor.

Let's see this for  $\text{Drinf}$ . The first thing to note is that  $\Gamma(\text{LocSys}^{\text{arithm}}, \mathcal{E}_{V,x} \otimes \text{Drinf}) : \mathbf{Vect} \rightarrow \mathbf{Vect}$  can be decomposed further by first unraveling the definition of  $\text{Drinf} : \mathbf{Vect} \rightarrow \text{QCoh}(\text{LocSys}^{\text{arithm}})$ .

**Notation.** In the following we will be abbreviating  $\text{Shv}(\text{Bun}_G)$  as  $D(\text{Bun}_G)$ , and will just write  $\text{LocSys}$  for the restricted version. This is mostly just so things have a remote chance of fitting on a page.

The idea is that if we have defined this via enhanced trace, we are realizing  $\text{QCoh}(\text{LocSys}^{\text{arithm}})$  as  $\text{Tr}(\text{Frob}^*, \text{QCoh}(\text{LocSys}))$ . That is, we write it as

$$\text{QCoh}(\text{LocSys}) \otimes_{\text{Frob}, \text{QCoh}(\text{LocSys} \times \text{LocSys}), \Delta} \text{QCoh}(\text{LocSys})$$

where the  $\text{Frob}$  means we use the graph of Frobenius for the map, and  $\Delta$  means we use the diagonal for the map to take the tensor product over.

Further breaking down the definition of  $\text{Drinf}$ , the enhanced trace constructs an element of

$$\text{Tr}(\text{Frob}^*, \text{QCoh}(\text{LocSys})) = \text{QCoh}(\text{LocSys}^{\text{arithm}}).$$

via the compatibility datum of  $\text{Frob}^!$  on  $\text{Shv}(\text{Bun}_G)$  as a module category over  $\text{QCoh}(\text{LocSys})$ . The following is simply unwinding the definition of this element from the compatibility datum.

The first functor we'll use to make  $\text{Drinf}$  is

$$\mathbf{Vect} \rightarrow D(\text{Bun}_G) \otimes_{\text{QCoh}(\text{LocSys})} D(\text{Bun}_G)$$

via the unit of the self-duality datum of  $D(\text{Bun}_G)$  as a module category over  $\text{QCoh}(\text{LocSys})$ .

Then we follow this with an isomorphism

$$D(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LocSys})} D(\mathrm{Bun}_G) \simeq (D(\mathrm{Bun}_G) \otimes D(\mathrm{Bun}_G)) \otimes_{\mathrm{QCoh}(\mathrm{LocSys} \times \mathrm{LocSys})} \mathrm{QCoh}(\mathrm{LocSys}).$$

There is a map

$$\Phi : (D(\mathrm{Bun}_G) \otimes D(\mathrm{Bun}_G)) \rightarrow \mathrm{QCoh}(\mathrm{LocSys}).$$

This is given by

$$\Gamma(\mathrm{LocSys}, \mathcal{E} \otimes \Phi(D_1, D_2)) := H^\bullet(\mathrm{Bun}_G, \mathrm{Frob}^!(D_1) \otimes^! (\mathcal{E} * D_2)).$$

Canonical self-duality on  $\mathrm{LocSys}^{\mathrm{restr}}$  means this actually makes sense as a map, which again formally follows from this being a reasonable stack.

Applying this map, we then have a map

$$\begin{array}{c} (D(\mathrm{Bun}_G) \otimes D(\mathrm{Bun}_G)) \otimes_{\mathrm{QCoh}(\mathrm{LocSys} \times \mathrm{LocSys})} \mathrm{QCoh}(\mathrm{LocSys}) \\ \downarrow \Phi \\ \mathrm{QCoh}(\mathrm{LocSys}) \otimes_{\mathrm{Frob}, \mathrm{QCoh}(\mathrm{LocSys} \times \mathrm{LocSys}), \Delta} \mathrm{QCoh}(\mathrm{LocSys}) \\ \downarrow \sim \\ \mathrm{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}}) \end{array}$$

To make the identification with  $\mathrm{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}})$  we have written it as  $\mathrm{Tr}(\mathrm{Frob}^*, \mathrm{QCoh}(\mathrm{LocSys}))$ : the last isomorphism is then using the definition of this trace via Hochschild homology.

This in total defines a functor

$$\mathbf{Vect} \rightarrow \mathrm{QCoh}(\mathrm{LocSys}^{\mathrm{arithm}})$$

matching **Drinf**. We can then tensor with  $\mathcal{E}_{V,x}$  and take global sections to get the overall result  $\Gamma(\mathrm{LocSys}^{\mathrm{arithm}}, \mathcal{E}_{V,x} \otimes \mathbf{Drinf}) : \mathbf{Vect} \rightarrow \mathbf{Vect}$  as a chain of simpler compositions.

Now note that starting at  $(D(\mathrm{Bun}_G) \otimes D(\mathrm{Bun}_G)) \otimes_{\mathrm{QCoh}(\mathrm{LocSys} \times \mathrm{LocSys})} \mathrm{QCoh}(\mathrm{LocSys})$  in this chain of compositions, we can replace the rest of the compositions by

$$\begin{array}{c} (D(\mathrm{Bun}_G) \otimes D(\mathrm{Bun}_G)) \otimes_{\mathrm{QCoh}(\mathrm{LocSys} \times \mathrm{LocSys})} \mathrm{QCoh}(\mathrm{LocSys}) \\ \downarrow \\ D(\mathrm{Bun}_G) \otimes D(\mathrm{Bun}_G) \\ \downarrow \Phi \\ \mathrm{QCoh}(\mathrm{LocSys}) \\ \downarrow \Gamma(\mathrm{LocSys}, \mathcal{E}_{V,x} \otimes -) \\ \mathbf{Vect} \end{array}$$

This helps: taking the full composition from  $\mathbf{Vect} \rightarrow D(\mathrm{Bun}_G) \otimes D(\mathrm{Bun}_G)$ , this now is the absolute unit of self-duality for  $\mathrm{Bun}_G$  and breaks down as

$$\mathbf{Vect} \longrightarrow D(\mathrm{Bun}_G) \xrightarrow{\Delta} D(\mathrm{Bun}_G \times \mathrm{Bun}_G) \simeq D(\mathrm{Bun}_G) \otimes D(\mathrm{Bun}_G).$$

where the first map corresponds to  $\omega_{\text{Bun}_G}$  (this is where  $\overline{\mathbf{Q}}_\ell$  is sent; it is the dualizing sheaf or unit of  $\otimes^!$  in the constructible setting). Thus,

$$\Gamma(\text{LocSys}^{\text{arithm}}, \mathcal{E}_{V,x} \otimes \text{Drinf}) : \mathbf{Vect} \rightarrow \mathbf{Vect}$$

can be identified with

$$\mathbf{Vect} \longrightarrow D(\text{Bun}_G) \xrightarrow{\Delta} D(\text{Bun}_G) \otimes D(\text{Bun}_G) \xrightarrow{\Phi} \mathbf{QCoh}(\text{LocSys}) \longrightarrow \mathbf{Vect}$$

where the last map is  $\Gamma(\text{LocSys}, \mathcal{E}_{V,x} \otimes -)$ , and we abuse notation with  $\Delta$  by using the isomorphism  $D(\text{Bun}_G \times \text{Bun}_G) \simeq D(\text{Bun}_G) \otimes D(\text{Bun}_G)$ .

In this, the only complicated map is  $\Phi$ . However, its defining property makes this far easier, as we have essentially given its definition by saying what it gives when composed with the map  $\Gamma(\text{LocSys}, \mathcal{E}_{V,x} \otimes -)$ . Applying this, the final component becomes

$$D(\text{Bun}_G) \otimes D(\text{Bun}_G) \xrightarrow{\text{Id} \otimes H_{V,x}} D(\text{Bun}_G) \otimes D(\text{Bun}_G) \xrightarrow{\text{Frob}^! \otimes \text{Id}} D(\text{Bun}_G) \otimes D(\text{Bun}_G)$$

followed by  $\Delta^!$ , which lands in  $D(\text{Bun}_G)$ , and then  $H^\bullet$  to get to  $\mathbf{Vect}$ . This is simply unwinding the definition of  $\Phi$ : the first part is using that  $\mathcal{E}_{V,x} * D_2$  is just the Hecke action, the second part gives us the result  $\text{Frob}^!(D_1)$ , and  $\Delta^!$  takes the shriek tensor product. Finally,  $H^\bullet$  completes the definition of  $\Phi$ .

At this point, we can now apply base change for the square giving the fiber product  $\text{Bun}_G \times_{\text{Frob}, \text{Bun}_G \times \text{Bun}_G} \text{Hecke}_x$ , now that we have reduced everything to

$$H^\bullet(\text{Bun}_G \times_{\text{Frob}, \text{Bun}_G \times \text{Bun}_G} \text{Hecke}_x, \mathcal{S}'_V) \simeq H^\bullet(\text{Bun}_G, \Delta^!(\text{Frob}^! \otimes H_{V,x})(\Delta_* \omega)).$$

Thus,  $\text{Drinf}$  satisfies the desired characterization.

We can see  $\mathcal{F} = \text{DrinfSht}$  also satisfies

$$\Gamma(\text{LocSys}^{\text{arithm}}, \mathcal{E}_{V,x} \otimes \mathcal{F}) \simeq H^\bullet(\text{Bun}_G \times_{\text{Frob}, \text{Bun}_G \times \text{Bun}_G} \text{Hecke}_x, \mathcal{S}'_V)$$

which is by definition in this setting. Thus, the two sheaves are isomorphic.

## REFERENCES

- [1] D. Gaitsgory, *From geometric to function-theoretic Langlands (or how to invent shtukas)*, arXiv:1606.09608.
- [2] C. Xue, *Smoothness of cohomology sheaves of stacks of shtukas*, arXiv:2012.12833.
- [3] V. Lafforgue, *Chtoucas pour les groupes réductifs et paramétrisation de Langlands globale*, J. Amer. Math. Soc. **31** (2018), no. 3, 719–891.